

Building My First Full-Stack Mini App

Study Tip Generator

Learn how React talks to Flask — one step at a time

React

Flask

JSON

What Are We Building?

A mini app where a student types their name, and gets a personalised study tip back!

1

Student enters name

React form with
an input field

2

Name sent to Flask

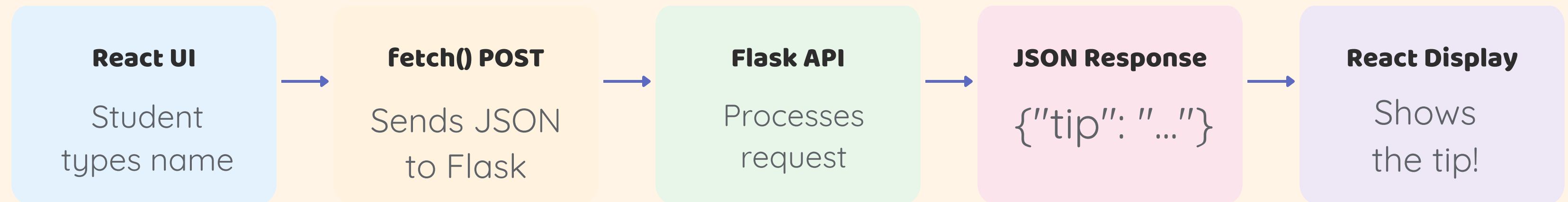
POST request to
/api/tip

3

Tip displayed

React shows the
JSON response

Full-Stack Architecture



The frontend and backend are separate apps that talk over HTTP

Project Setup

Tools you need: Python 3, Node.js, a code editor (VS Code)

```
my-app/  
├── backend/  
│   ├── app.py  
│   └── requirements.txt  
└── frontend/  
    ├── src/App.js  
    └── package.json
```

```
# Backend setup  
cd backend  
pip install flask flask-cors  
  
# Frontend setup  
cd frontend  
npx create-react-app .  
npm start
```

Step 1: Set Up Flask

Flask is a lightweight Python web framework. We use it to create an API endpoint.

Route

A URL path like /api/tip that handles requests

POST method

Receives data (the student name) from the frontend

jsonify()

Converts Python dict to JSON for the response

flask-cors allows React (on port 3000) to talk to Flask (on port 5000)

app.py — Flask Backend

```
# File: backend/app.py

from flask import Flask, request, jsonify
from flask_cors import CORS

app = Flask(__name__)
CORS(app)

@app.route('/api/tip', methods=['POST'])
def get_tip():
    data = request.get_json()
    name = data['name']
    tip = f"Hey {name}, try the Pomodoro method today!"
    return jsonify({'tip': tip})

if __name__ == '__main__':
    app.run(debug=True)
```

Key points:

- POST receives JSON body
- jsonify() sends JSON back
- CORS allows cross-origin

Test Your Backend

Run the server, then send a test request

```
$ python app.py  
* Running on http://127.0.0.1:5000
```

Example Request (using curl or Postman):

```
curl -X POST http://127.0.0.1:5000/api/tip \  
  -H "Content-Type: application/json" \  
  -d '{"name": "Asha"}'
```

Response:

```
{ "tip": "Hey Asha, try the Pomodoro method  
today!" }
```

200 OK ✓

Step 2: Set Up React

Now we build the frontend — a simple form that collects a student name and sends it to Flask.

 frontend/src/App.js

What the React app does:

1. Shows an input box where the student types their name
2. Has a Submit button to send the name
3. Sends the name to Flask using a POST request
4. Displays the study tip Flask sends back

Form Preview

Get My Tip

Your tip appears here!

App.js — React Frontend

This component handles the form, sends data to Flask, and shows the tip.

```
const [name, setName] = useState('')
const [tip, setTip] = useState('')

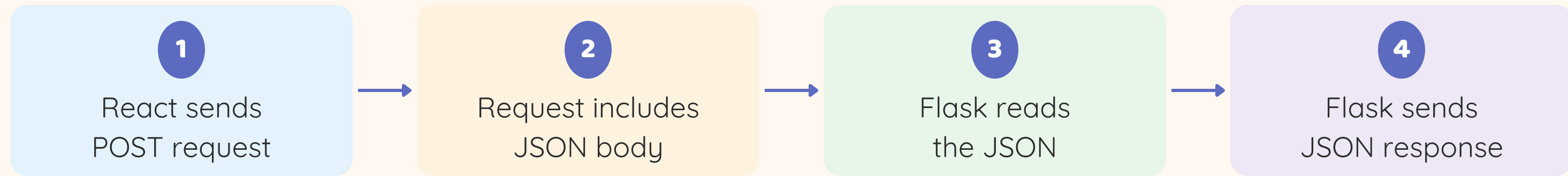
const handleSubmit = async (e) => {
  e.preventDefault()
  const response = await fetch(
    'http://127.0.0.1:5000/api/tip', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ name })
  })
  const data = await response.json()
  setTip(data.tip)
}
```

Key parts:

- useState — stores the name and tip
- fetch() — sends POST to Flask
- data.tip — shows the response

Connecting React to Flask

Here's how data flows from the browser to the server and back.



Endpoint: `/api/tip` — this is the URL React sends data to

Header: `Content-Type: application/json` — tells Flask it's JSON

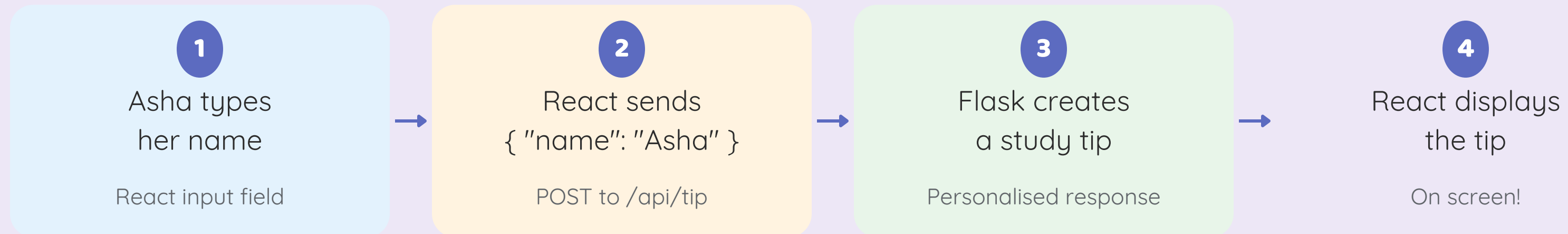
Body: `{ "name": "Asha" }` — the data React sends

CORS: Flask needs flask-cors so the browser allows the request

Without CORS, the browser blocks requests between different ports (3000 → 5000).

End-to-End Example

Follow one complete journey — from typing a name to seeing the tip.



Sample Response from Flask:

```
{"tip": "Asha, revise for 25 minutes and take a 5-minute break."}
```

React receives this JSON and shows the tip text to Asha on screen.

Troubleshooting + Next Steps

Common Issues

✗ Flask server not running

Make sure you ran `python app.py` first

✗ Wrong endpoint URL

Check the port and path: `/api/tip` on 5000

✗ CORS error in console

Add `flask-cors` to your Flask app

Next Steps

✓ Add a loading state

Show 'Loading...' while waiting for Flask

✓ Tips for different subjects

Send a subject field and customise tips

✓ Add a class or goal field

Let students pick their class or study goal

Every bug is a learning moment. You're doing great! 💪

You Built a Full-Stack Mini App! 🎉

React collects the name → Flask creates the study tip → React shows the result

React

Flask

JSON

Keep building. Keep learning. You've got this! 🚀